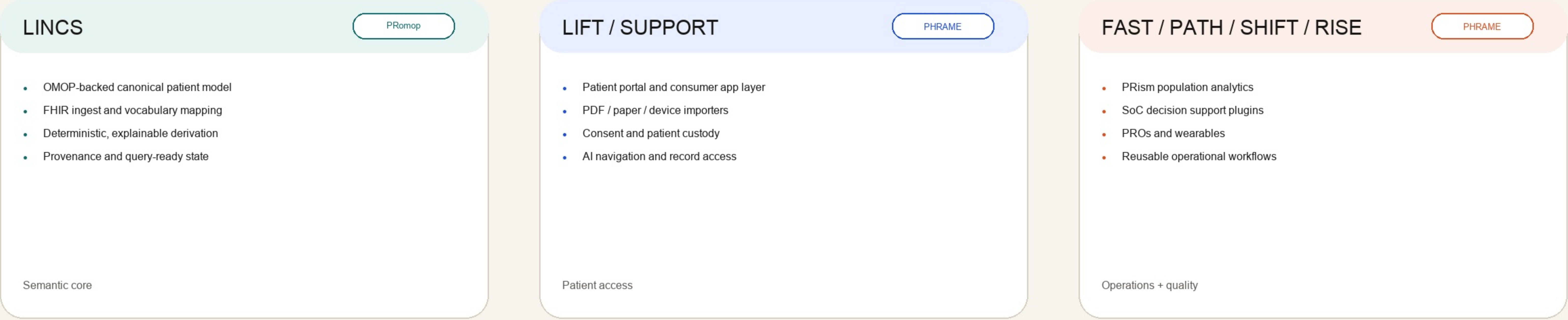


Executive Summary

Utah's RHTP asks for a statewide, cloud-based, semantically interoperable data platform. PRomop already gives Utah the semantic core of that platform: a canonical PatientRecord derived from OMOP and fed by FHIR, with terminology mapping, provenance, and query-ready patient state. PHRAME then extends the stack upward into patient custody, importers, decision support, trial matching, telehealth-adjacent workflows, and population analytics.

Recommended position

- Use PRomop as the semantic backbone for LINC3 because it already harmonizes heterogeneous records into one computable patient contract.
- Use PHRAME as the broader operational envelope for the rest of RHTP because it adds patient-custodied workflows, importers, AI-enabled navigation, and reusable applications.
- Treat the combination as a single architecture with distinct layers: PRomop for meaning, PHRAME for delivery.



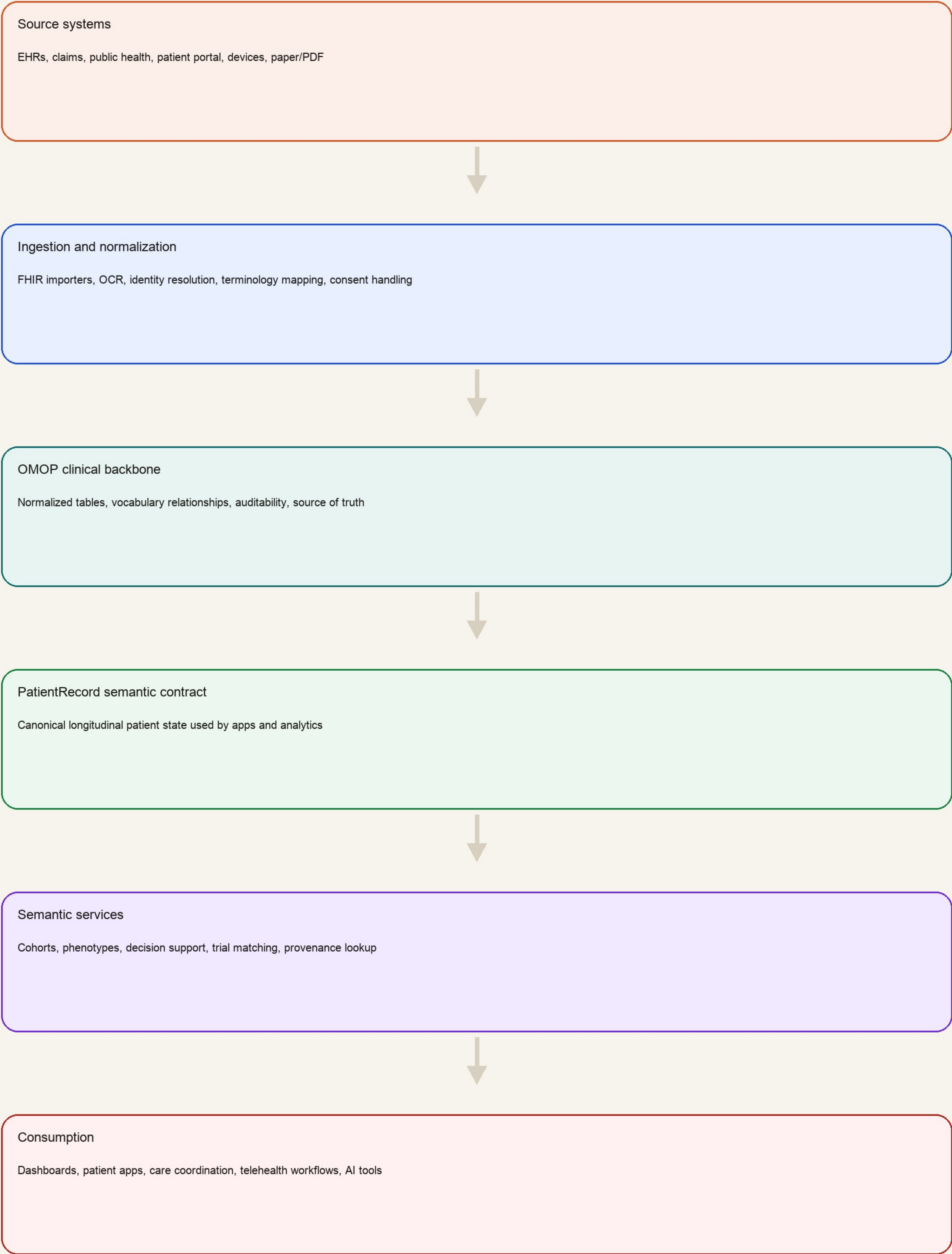
Bottom line

PRomop is the strongest fit for Utah LINC3 because it already does the hardest semantic work: take messy source data, normalize it, and materialize one shared patient model. PHRAME is the right complementary layer because it turns that model into a patient-facing, multi-application platform for the rest of the transformation plan.

Architecture Overview

PRomop is the semantic core; PHRAME is the service envelope.

Layered model for Utah RHTP



Why this structure matters

- It keeps meaning stable: source systems can vary without forcing downstream apps to relearn every schema.
- It lets Utah add claims, registry, and community data later without breaking the patient contract.
- It gives the state one place to explain where a value came from and how it was derived.
- It separates what is standardized (OMOP and concepts) from what is reusable (PatientRecord and semantic services).

PRomop already provides

- FHIR R4 ingestion and import pipelines
- OMOP tables as the clinical system of record
- Concept mapping and hierarchy support
- PatientRecord derivation for decision-ready access
- Provenance as a first-class feature

PHRAME adds

- Patient custodied portals and navigation
- PDF / paper / device / PRO importers
- EXACT for trial matching
- SoC Service for disease-specific CDS
- PRism for population analytics

PRomop for LINC

This is the piece Utah should treat as the semantic backbone of the statewide data platform.

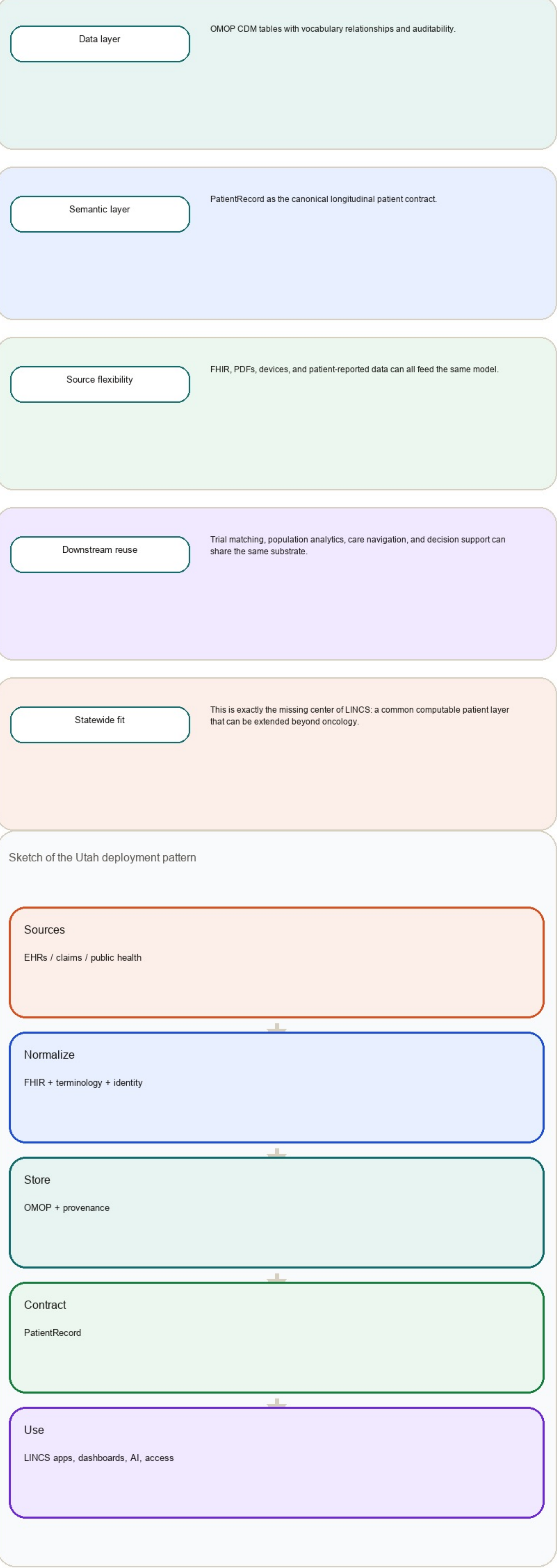
What PRomop already solves

- Canonical patient representation. PatientRecord consolidates longitudinal facts into one reusable row-level contract for downstream consumers.
- Terminology normalization. FHIR resources are mapped into OMOP tables and standard concepts, with concept sets represented in OMOP Concept / relationship tables and HemOnc-derived vocabularies where appropriate.
- Explainable derivation. The data model is deterministic and re-derived from OMOP, so the meaning of a field can be traced back to source data and transformations.
- Cross-source harmonization. PRomop is designed to accept data from multiple EHRs and normalize it into one patient view, which is exactly what LINC needs statewide.
- Research and operational reuse. The same semantic model can power trial matching, public-health measures, care coordination, and AI features without forcing each use case to rebuild its own patient state.

How this maps to Utah LINC

- The Utah RHTP explicitly asks for a cloud-based, semantically interoperable data platform that harmonizes data from EHRs, claims, and public health systems into a single computable structure and exposes standardized data through FHIR APIs.
- PRomop is the concrete implementation pattern for that requirement: normalize once, model once, and reuse everywhere.
- If Utah uses PRomop as the core semantic layer, LINC becomes a statewide expansion project rather than a greenfield build.

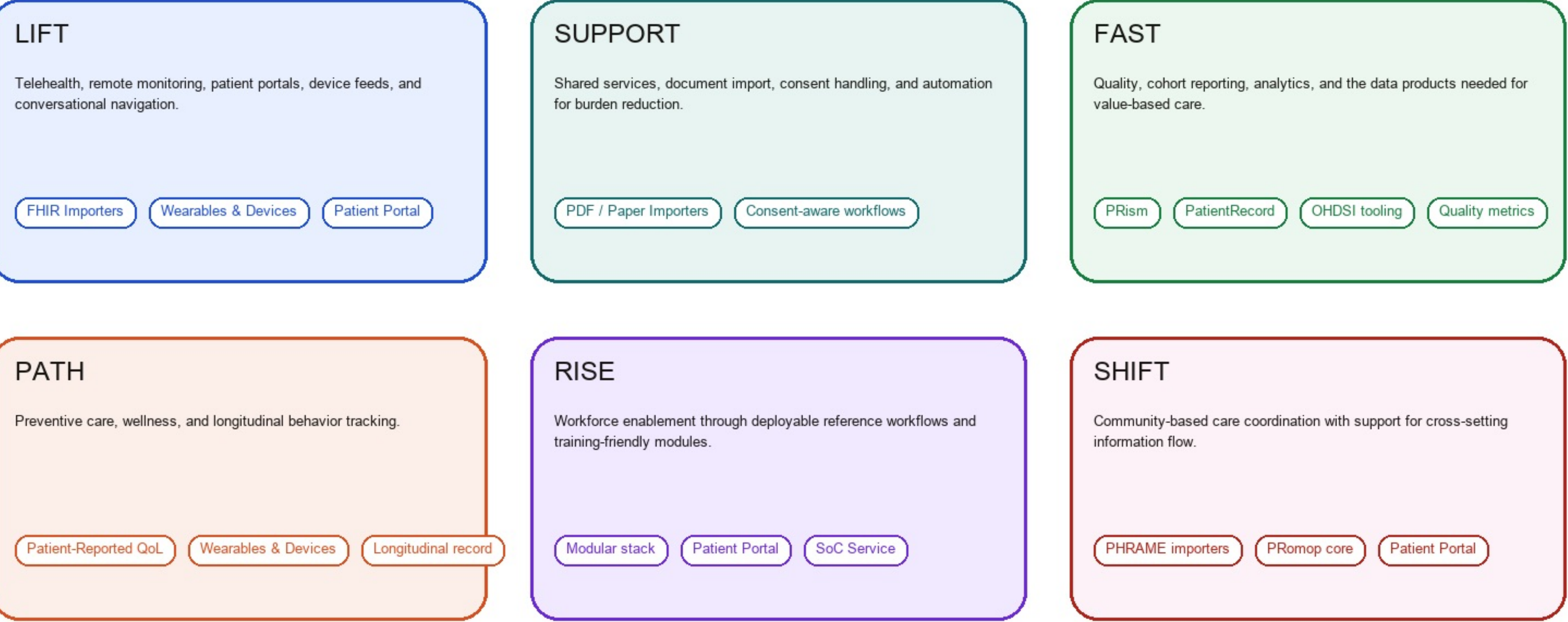
Capability matrix



PHRAME for the rest of RHTP

PHRAME extends the platform where Utah needs patient access, operational tooling, and reusable applications.

RHTP initiative alignment



Why PHRAME matters to Utah beyond LINCS

- It gives Utah a patient-custodied application layer rather than only a data platform.
- It comes with modular components already mapped to the problems RHTP keeps naming: patient access, AI navigation, document ingestion, analytics, and trial matching.
- It is open-source and modular, which matters if Utah wants to own and adapt the stack rather than license a closed platform.
- It has already been validated in oncology operations at real scale, which lowers implementation risk for the state.

Operational interpretation

Utah should think of PHRAME as the application and workflow layer that sits on top of the semantic core. Where PRomop says "this is what the patient record means," PHRAME says "this is how patients, coordinators, and clinicians use it." That distinction lets Utah reuse one data backbone across many RHTP initiatives without forcing every initiative to invent its own record model.

Recommended Adoption Plan

Use PRomop as the semantic core and PHRAME as the reusable service envelope.

Phased implementation

Phase 1

Lock the semantic contract

Define PatientRecord and its statewide extensions as the canonical patient model for LINCS.

Phase 2

Expand source coverage

Add claims, public-health registry feeds, patient-generated data, and community-partner feeds.

Phase 3

Expose semantic services

Add value sets, cohort and phenotype services, provenance lookup, and API endpoints.

Phase 4

Operationalize statewide use

Stand up dashboards, patient access, care navigation, and shared workflow services.

Decision criteria for Utah

Ownability

Can the state deploy, extend, and govern it without a vendor lock-in penalty?

Semantic depth

Does it already preserve meaning at the patient level, not just at the transport layer?

Workflow breadth

Does it support the full range of RHTP use cases: access, analytics, CDS, and navigation?

Adoption risk

Can Utah start with the semantic core and phase in the rest without a big-bang replacement?

Selected links

- PHRAME why: <https://phrame.org/#why>
- PHRAME architecture: <https://phrame.org/#architecture>
- PHRAME compare: <https://phrame.org/#compare>
- PHRAME cases: <https://phrame.org/#cases>
- PHRAME engage: <https://phrame.org/#engage>
- HealthKey GitHub org: <https://github.com/healthkey-ai>
- Repos: [promop](#), [FHIR_importers](#), [EXACT](#), [SoC](#), [prism](#)

Recommendation

Use PRomop for LINCS and PHRAME for the rest of the service layer. That gives Utah a statewide semantic core plus a practical patient-facing platform.